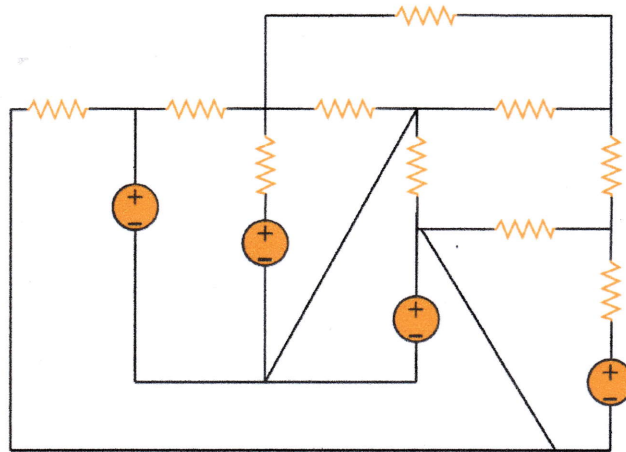


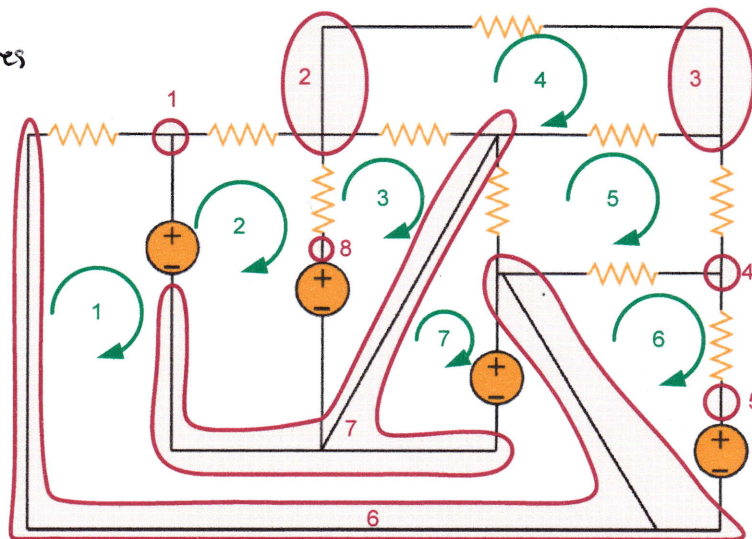
Topic 2: Kirchhoff's Laws, Nodal & Mesh Analysis

1. Identify all the nodes, branches, and meshes in the following figure.



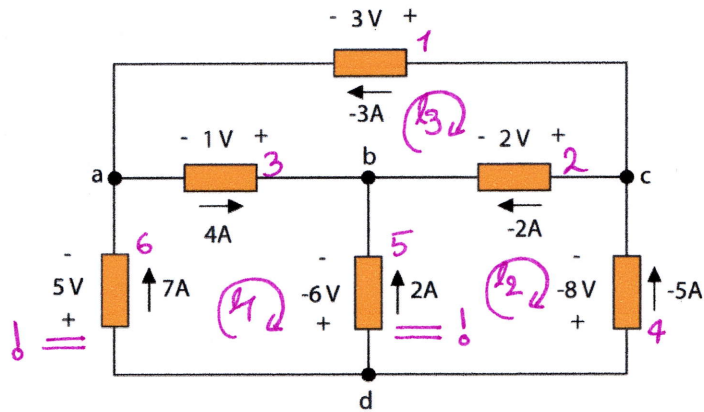
Solution:

{ 14 branches
 8 nodes
 7 meshes



2. In the following circuit (Question 2 of the Online Tutorial 1) the total sum of powers being consumed and generated is not equal to zero, violating the conservation of energy law.

- Identify the error in the circuit using Kirchhoff's Current and Voltage Laws.
- Confirm your answer by calculating the powers and their total sum.



Solution: First check KCLs and KVLs

KCL: @ node a: $7 - 3 - 4 = 0 \checkmark$

@ node b: $4 + 2 - 2 = 4 \neq 0 \times$ error

@ node c: $-5 + 2 + 3 = 0 \checkmark$

@ node d: $7 + 2 - 5 = 4 \neq 0 \times$ error

This suggests branch 5 has the wrong direction: $\downarrow 2A \checkmark$ or $\uparrow -2A$

KVL: in l_1 : $+5 - 1 - (-6) = 10 \neq 0 \times$ error

in l_2 : $+(-6) - 2 - (-8) = 0 \checkmark$

in l_3 : $+(-3) + 2 + 1 = 0 \checkmark$

This suggests branch 6 has the wrong polarity: $\underline{+5V}$ or $\underline{-5V}$

Summation of powers with corrected direction and polarity:

$$P_1 = -9W \quad P_4 = 40W$$

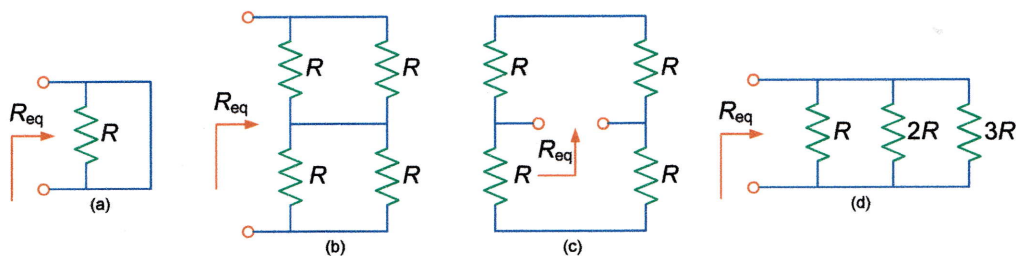
$$P_2 = -4W \quad P_5 = 12W$$

$$P_3 = -4W \quad P_6 = -35W$$

$$\Rightarrow \sum_{n=1}^6 P_n = 40 + 12 - 35 - 4 - 4 - 9 = 0 \checkmark$$

Confirmed

3. Find the equivalent resistance of the following networks.



Solution: a) Extreme case of short circuit being parallel with $R \Rightarrow R_{eq} = 0\Omega$

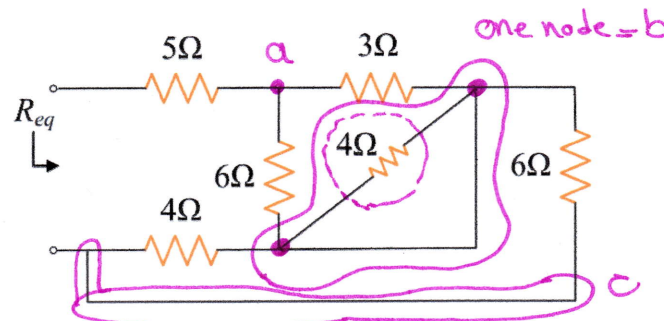
$$b) \quad R \parallel R = \frac{R}{2} \quad \Rightarrow \quad (R \parallel R) + (R \parallel R) \Rightarrow R_{eq} = \frac{R^2}{2R} + \frac{R^2}{2R} = R$$

$$c) \quad 2R \parallel 2R \Rightarrow R_{eq} = \frac{2R \times 2R}{4R} = R$$

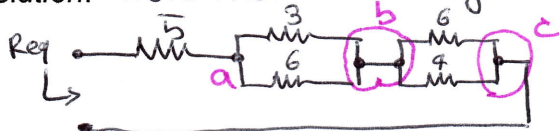
$$d) \quad R_{eq} = R \parallel 2R \parallel 3R \Rightarrow \frac{1}{R_{eq}} = \frac{1}{R} + \frac{1}{2R} + \frac{1}{3R} = \frac{6R^2 + 3R^2 + 2R^2}{6R^3} = \frac{11R^2}{6R^3}$$

$$\Rightarrow R_{eq} = \frac{6}{11} R$$

4. (Midterm Exam - S1, 2016) For the circuit below, calculate the equivalent resistance R_{eq} of the network as seen from the terminals.

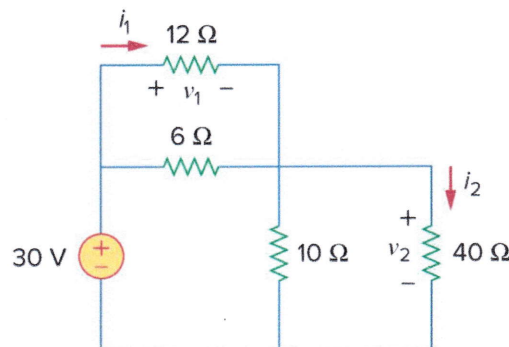


Solution: Note that 4Ω is being short circuited.



$$R_{eq} = 5 + 3 \parallel 6 + 6 \parallel 4 = 5 + \frac{3 \times 6}{3 + 6} + \frac{6 \times 4}{6 + 4} = 5 + 2 + 2.4 = 9.4 \Omega$$

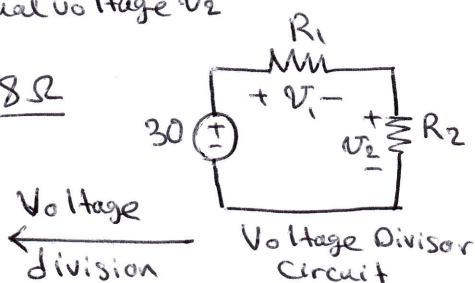
5. Find v_1 and v_2 in the circuit shown below. Also the currents i_1 and i_2 .



Solution: Note that $12\Omega \parallel 6\Omega$ with equal voltage v_1 and $10\Omega \parallel 40\Omega$ with equal voltage v_2

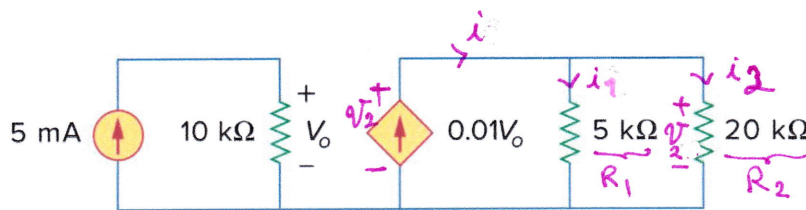
$$R_1 = \frac{12 \times 6}{12 + 6} = 4\Omega, R_2 = \frac{10 \times 40}{10 + 40} = 8\Omega$$

$$\begin{cases} v_1 = \frac{R_1}{R_1 + R_2} \times 30 = \frac{4 \times 30}{4 + 8} = 10V \\ v_2 = \frac{R_2}{R_1 + R_2} \times 30 = \frac{8 \times 30}{4 + 8} = 20V \end{cases}$$



$$\text{Ohm's law: } i_1 = \frac{v_1}{12\Omega} = \frac{10}{12} = \frac{5}{6} \approx 833.3 \text{ mA}, i_2 = \frac{v_2}{40} = \frac{20}{40} = 0.5 \text{ A} = 500 \text{ mA}$$

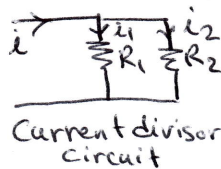
6. In the following circuit, calculate the current and voltage of the $20 \text{ k}\Omega$ resistor. What is the power generated/supplied by the dependent current source?



Solution: Note that 5mA only flows through 10K Ω

$$\rightarrow V_0 = 10K \times 5mA = \underline{50V}$$

Voltage-Controlled Current Source $\rightarrow i = 0.01V_0 = 0.01 \times 50 = \underline{0.5A}$



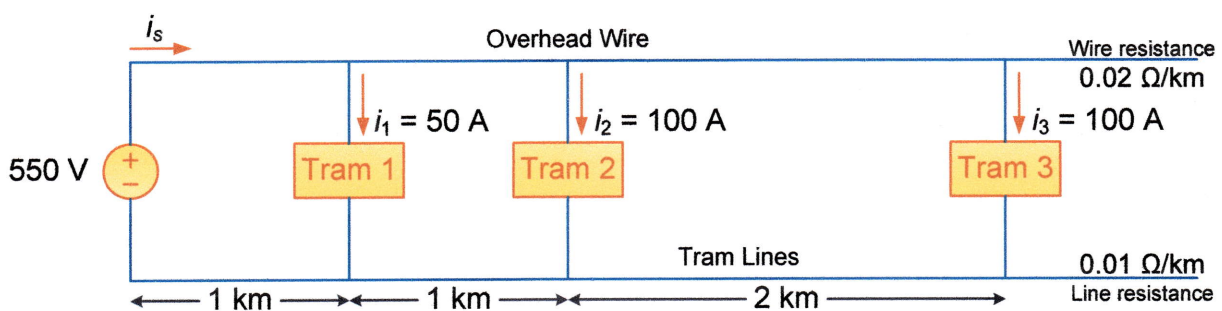
$$\rightarrow i_2 = \frac{R_1}{R_1 + R_2} \times i = \frac{5K}{5K + 20K} \times 0.5 = \underline{0.1A}$$

$$V_2 = 20K \times i_2 = 20K \cdot 0.1A = 2KV = \underline{2000V}$$

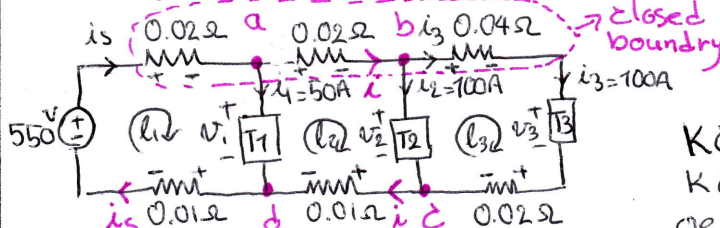
VCCS has the same voltage as $V_2 \rightarrow P_{VCCS} = V_2 \times i = 2K \times 0.5A = \underline{1KW}$

7. The DC traction power supply system (TPSS) of a light rail network, as shown in the following figure, consists of a DC Voltage source, overhead wires and an earth return. At a given moment, three trams travel within one section post and absorb 50 A, 100 A, and 100 A, respectively. Considering the overhead wires electric resistance given in units of Ω/km , calculate the power that is absorbed by each trams. What is the overall efficiency of this TPSS?

Efficiency η is defined as the ratio of the total **useful power** consumed by a system (to perform a task) to the **total power supplied** to it. Efficiency is an indication of how much of the supplied power is being lost like the heat dissipation by wires resistances.



Solution: Draw the equivalent circuit. Note that wires and lines can be modelled with their equivalent resistance according to their length in the circuit diagram.



KCL @ b: $i = i_2 + i_3 = 200A \triangle @ c$
 KCL @ a: $i_s = i + i_1 = 250A \triangle @ d$
 OR you can consider a closed boundary with resistors inside

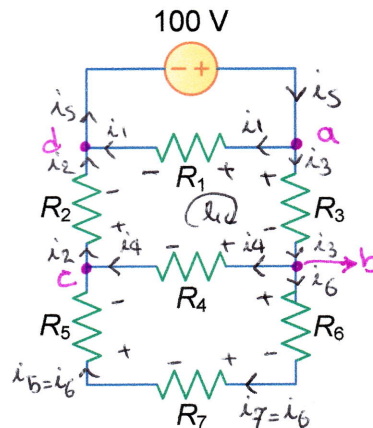
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$$\begin{aligned}
 \text{KVL in } l_1: -550 + 0.02 i_5 + V_1 + 0.01 i_5 &= 0 \Rightarrow V_1 = 550 - 0.03 \times 250 = \underline{542.5 \text{ V}} \\
 \text{KVL in } l_2: -V_1 + 0.02 i_1 + V_2 + 0.01 i_1 &= 0 \Rightarrow V_2 = 542.5 - 0.03 \times 200 = \underline{536.5 \text{ V}} \\
 \text{KVL in } l_3: -V_2 + 0.04 i_3 + V_3 + 0.02 i_3 &= 0 \Rightarrow V_3 = 536.5 - 0.06 \times 100 = \underline{530.5 \text{ V}} \\
 P_1 = V_1 \cdot i_1 = 542.5 \times 50 &= 27.125 \text{ kW} \\
 P_2 = V_2 \cdot i_2 = 536.5 \times 100 &= 53.65 \text{ kW} \\
 P_3 = V_3 \cdot i_3 = 530.5 \times 100 &= 53.05 \text{ kW} \\
 \Rightarrow \eta = \frac{P_1 + P_2 + P_3}{P_s} = \frac{133.825 \text{ kW}}{137.5 \text{ kW}} &= 0.973 = \underline{97.3\%}
 \end{aligned}$$

8. An industrial heating application can be modelled by using a connection of resistive elements as shown in the figure below. The two requirements for this industrial heating application are stated as below,

- 1) The total power consumed is equal to 3500 W.
- 2) The elements generate heat uniformly (this means that they dissipate equal power).

Based on the above requirements, calculate the value of resistances R_1 to R_7 .



Solution: Since each resistor consumes the same amount of power, thus the power of each resistor is $\frac{3500}{7} = \underline{500 \text{ W}} = P_1 = P_2 = P_3 = P_4 = P_5 = P_6 = P_7 = P$

R_1 is in parallel with 100V $\Rightarrow V_{R_1} = 100 \text{ V} \Rightarrow P = \frac{V_{R_1}^2}{R_1} \Rightarrow R_1 = \frac{V_{R_1}^2}{P} = \frac{10000}{500} = \underline{20 \Omega}$

Also $i_1 = \frac{V_{R_1}}{R_1} = \frac{100}{20} = \underline{5 \text{ A}}$

Note that Total power absorbed = total power supplied. Thus the 100V source supplies 3500W

$\Rightarrow i_5 = \frac{3500 \text{ W}}{100 \text{ V}} = \underline{35 \text{ A}}$

KCL @ b & c: $\begin{cases} i_3 = i_4 + i_6 \\ i_2 = i_4 + i_6 \end{cases} \Rightarrow i_2 = i_3$. Also $R_5 = R_6 = R_7$ since they have the same current and consume the same power $\Rightarrow R_2 = R_3$ as they also consume the same power

KCL @ a $i_5 = i_1 + i_3 \Rightarrow i_3 = 35 - 5 = \underline{30 \text{ A}}$

Now $V_{R_3} = V_{R_2}$ since $R_2 = R_3$ and $i_2 = i_3 \Rightarrow V_{R_2} = \frac{500}{\frac{1}{2} \times 30} = \underline{\frac{50}{3} \text{ V} = V_{R_3}}$

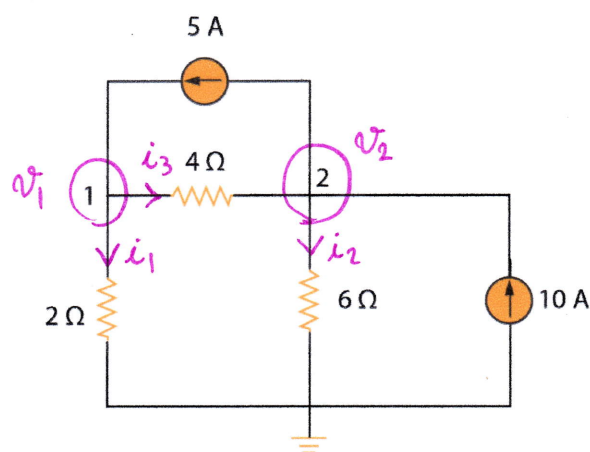
KVL $l_1: -V_{R_1} + V_{R_2} + V_{R_4} + V_{R_3} = 0 \Rightarrow V_{R_4} = 100 - 2 \times \frac{50}{3} = \underline{\frac{200}{3} \text{ V}}$

$\Rightarrow P = \frac{V_{R_4}^2}{R_4} \Rightarrow R_4 = \frac{(\frac{200}{3})^2}{500} = \underline{\frac{400}{9} \Omega}$

$i_6 = i_3 - i_4 = 30 - \frac{200}{\frac{400}{9}} = 30 - \frac{15}{2} = \underline{\frac{45}{2} \text{ A}}$

$P = R_6 i_6^2 \Rightarrow R_6 = \frac{500}{(\frac{45}{2})^2} = \frac{2000}{2025} = \underline{0.987 \Omega} = R_5 = R_7$

9. Find the node voltages in the circuit given below.



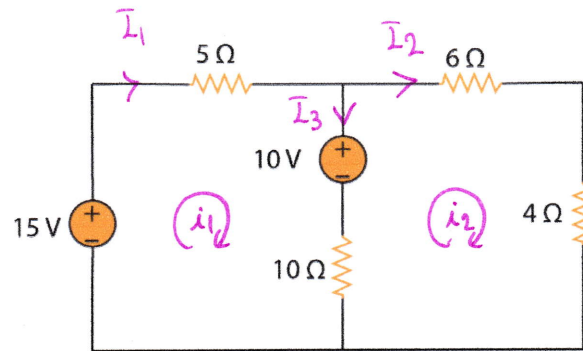
Answer: $v_1 = \frac{40}{3} = 13.333 \text{ V}$ and $v_2 = 20 \text{ V}$.

Solution: Assign v_1 and v_2 to nodes 1 and 2 and choose arbitrary direction for currents.

$$\begin{aligned} \text{KCL @ } n1: 5 &= \frac{v_1}{2} + \frac{v_1 - v_2}{4} \quad \times 4 \rightarrow 20 = 3v_1 - v_2 \\ \text{KCL @ } n2: 10 + \frac{v_1 - v_2}{4} &= 5 + \frac{v_2}{6} \quad \times 12 \rightarrow 60 = 5v_2 - 3v_1 \Rightarrow \begin{cases} 3v_1 - v_2 = 20 \\ 3v_1 - 5v_2 = -60 \end{cases} \\ \Rightarrow v_1 &= \frac{40}{3} = 13.333 \text{ V} \\ v_2 &= 20 \text{ V} \end{aligned}$$

Note that in the case of $i_3 = \frac{v_1 - v_2}{4} = -1.66 \text{ A}$ showing that the actual direction of current is opposite.

10. In the following circuit, find all the branch currents using mesh analysis. (Assign mesh currents and branch currents individually)



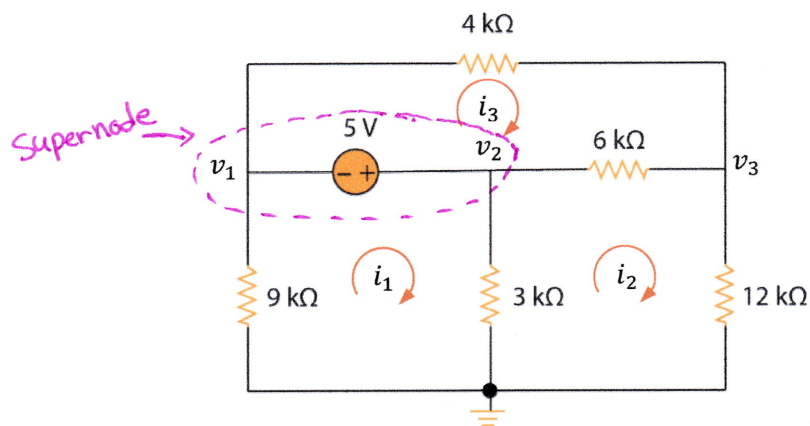
Answer: $I_1 = 1$ A (current in 15-V voltage source and 5-Ω resistor), $I_2 = 1$ A (current in 6-Ω and 4-Ω resistors), and $I_3 = 0$ A (current in 10-V voltage source and 10-Ω resistor).

Solution: KVL in m1: $-15 + 5i_1 + 10 + 10(i_1 - i_2) = 0$
 KVL in m2: $6i_2 + 4i_2 - 10(i_1 - i_2) - 10 = 0$

$$\Rightarrow \begin{cases} 15i_1 - 10i_2 = 5 & \times \frac{1}{5} \\ -10i_1 + 20i_2 = 10 & \times -\frac{1}{10} \end{cases} \Rightarrow \begin{cases} 3i_1 - 2i_2 = 1 \\ i_1 - 2i_2 = -1 \end{cases} \Rightarrow \begin{cases} i_1 = 1\text{A} \\ i_2 = 1\text{A} \end{cases}$$

Thus: $\begin{cases} I_1 = i_1 = 1\text{A} \\ I_2 = i_2 = 1\text{A} \\ I_3 = i_1 - i_2 = 0\text{A} \end{cases}$

11. For the circuit below,
 a) Write nodal equations using nodal analysis
 b) Write mesh equations using mesh analysis



Answer:

a) $\begin{cases} v_1 - v_2 = -5 \\ 13v_1 + 18v_2 - 15v_3 = 0 \\ 3v_1 + 2v_2 - 6v_3 = 0 \end{cases}$

b) $\begin{cases} 12i_1 - 3i_2 = 5 \\ 3i_1 - 21i_2 + 6i_3 = 0 \\ 6i_2 - 10i_3 = 5 \end{cases}$

Solution: Nodal analysis

* KCL @ V_1 : $\frac{V_1}{2} + \frac{V_2}{3} + \frac{V_2 - V_3}{6} + \frac{V_1 - V_3}{4} = 0$
 Supernode $\xrightarrow{\times 36} 13V_1 + 18V_2 - 15V_3 = 0$

* KCL @ V_3 : $\frac{V_3}{12} = \frac{V_2 - V_3}{6} + \frac{V_1 - V_3}{4}$
 $\xrightarrow{\times 12} 3V_1 + 2V_2 - 6V_3 = 0$

* Supernode Constraint (KVL) $V_2 - V_1 = 5$
 $\Rightarrow \begin{cases} V_1 - V_2 = -5 \\ 13V_1 + 18V_2 - 15V_3 = 0 \\ 3V_1 + 2V_2 - 6V_3 = 0 \end{cases}$ Nodal equation

Mesh analysis

* KVL in m_1 : $9i_1 - 5 + 3(i_1 - i_2) = 0$
 $\rightarrow 12i_1 - 3i_2 = 5$

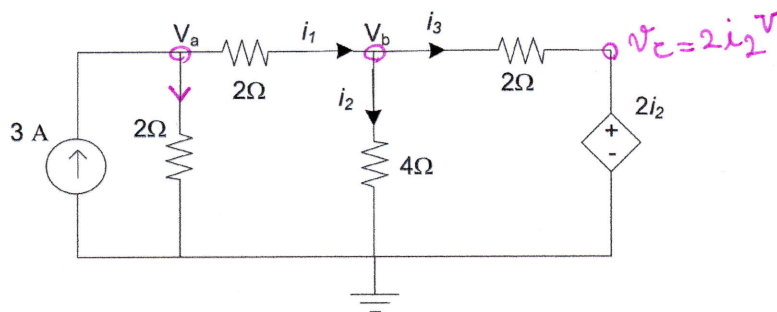
* KVL in m_2 : $12i_2 - 3(i_1 - i_2) + 6(i_2 - i_3) = 0$
 $\rightarrow -3i_1 + 21i_2 - 6i_3 = 0$ $\xrightarrow{\times -1}$

* KVL in m_3 : $4i_3 - 6(i_2 - i_3) + 5 = 0$
 $\rightarrow 10i_3 - 6i_2 = -5$ $\xrightarrow{\times -1}$

$\Rightarrow \begin{cases} 12i_1 - 3i_2 = 5 \\ 3i_1 - 21i_2 - 6i_3 = 0 \\ 6i_2 - 10i_3 = 5 \end{cases}$ Mesh equation

12. (Mid-session Exam – S2, 2016) For the circuit below,

- Apply nodal analysis to write down the node voltage equations at nodes V_a and V_b .
- Solve the voltage equations in part (a) to find the voltages V_a and V_b .
- Find the currents i_1 , i_2 and i_3 based on your results in part (b)



Answer:

- $\begin{cases} 2v_a - v_b = 6 \\ 2v_a - 4v_b = 0 \end{cases}$
- $V_a = 4\text{ V}$ and $V_b = 2\text{ V}$.
- $i_1 = 1\text{ A}$, $i_2 = 0.5\text{ A}$ and $i_3 = 0.5\text{ A}$

Solution: Note that V_c is directly connected to dependent voltage source

a) * KCL @ V_a : $3 = \frac{V_a}{2} + \frac{V_a - V_b}{2} \xrightarrow{\times 2} 2V_a - V_b = 6$

* KCL @ V_b : $\frac{V_a - V_b}{2} = \frac{V_b}{4} + \frac{V_b - V_c}{2}$ Also $V_c = 2i_2$ and $i_2 = \frac{V_b}{4} \Rightarrow V_c = \frac{V_b}{2}$ I

$\xrightarrow{\times 4 \& \text{I}} 2(V_a - V_b) = V_b + 2(V_b - \frac{V_b}{2}) \rightarrow 2V_a - 4V_b = 0$

$\Rightarrow$ Nodal equation $\begin{cases} 2V_a - V_b = 6 \\ 2V_a - 4V_b = 0 \end{cases}$

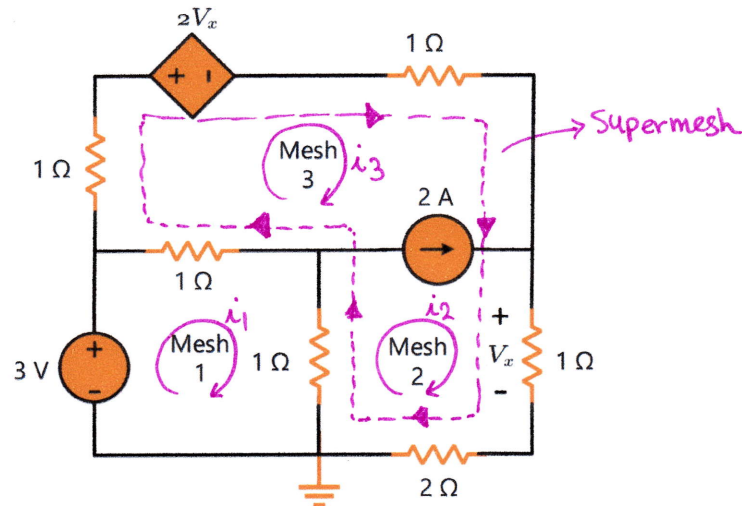
b) Solve the equations $\rightarrow V_a = 4\text{ V}$ and $V_b = 2\text{ V}$

c) $i_1 = \frac{V_a - V_b}{2} = \frac{4 - 2}{2} = 1\text{ A}$

$i_2 = \frac{V_b}{4} = \frac{2}{4} = 0.5\text{ A}$

$i_3 = \frac{V_b - V_c}{2} \xrightarrow{\text{I}} \frac{V_b - \frac{V_b}{2}}{2} = \frac{V_b}{4} = \frac{2}{4} = \frac{1}{2} \text{ or } 0.5\text{ A}$

13. (Mid-session Exam – Summer, 2017) For the circuit below,
- Apply mesh analysis to write down the mesh current equations in Mesh 1, Mesh 2, and Mesh 3 (this part was only given in the actual exam).
 - Solve the current equations in part (a) to find the mesh current.
 - Find the power generated by the dependent voltage source based on your results in part (b).



Answer:

- $$\begin{cases} 2i_1 - i_2 - i_3 = 3 \\ 2i_1 - 6i_2 - 3i_3 = 0 \\ i_2 - i_3 = 2 \end{cases}$$
- $i_1 = 1.5 \text{ A}, i_2 = 1 \text{ A} \text{ and } i_3 = -1 \text{ A}$
- $P = 2 \text{ W}$ supplied

Solution: Note that 2 A current source is a common branch between Mesh 2 and Mesh 3.

a) * KVL in m1: $-3 + 1(i_1 - i_3) + 1(i_1 - i_2) = 0 \rightarrow 2i_1 - i_2 - i_3 = 3$

* KVL in Supermesh: $1i_3 + 2V_x + 1i_3 + 1i_2 - 1(i_1 - i_2) - 1(i_1 - i_3) = 0$

$\rightarrow 2V_x - 2i_1 + 4i_2 + 3i_3 = 0$ Also $V_x = 1i_2$ (Note the polarity of V_x and direction of i_2)

$\Rightarrow -2i_1 + 6i_2 + 3i_3 = 0$ $\xrightarrow{\times -1}$

* Supermesh Constraint (KCL): $i_2 - i_3 = 2$

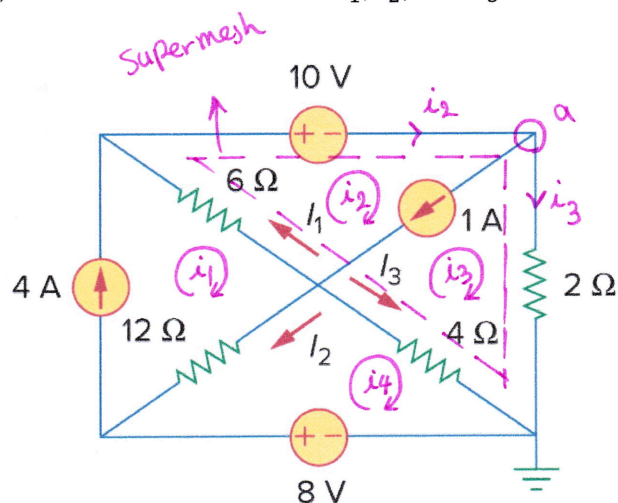
Thus:
$$\begin{cases} 2i_1 - i_2 - i_3 = 3 \\ 2i_1 - 6i_2 - 3i_3 = 0 \\ i_2 - i_3 = 2 \end{cases}$$

b) Solve the equations $\rightarrow \begin{cases} i_1 = 1.5 \text{ A} \\ i_2 = 1 \text{ A} \\ i_3 = -1 \text{ A} \end{cases}$

c) $P = -(-1) \times 2V_x \stackrel{\textcircled{I}}{=} 1 \times 2 \times 1 = 2 \text{ W}$ supplied or -2W absorbed

Note that i_3 enters the positive terminal of the dependent voltage source.

14. In the circuit below, find the branch currents I_1 , I_2 , and I_3 .



Answer: $I_1 = -1\text{ A}$, $I_2 = 0\text{ A}$, and $I_3 = 2\text{ A}$

Solution: Note that 1A current source is a common branch between mesh 2 and mesh 3.

Also, it is clear that $i_1 = 4\text{ A}$ (I) in mesh 1

* KVL in m4: $-8 - 12(i_1 - i_4) + 4(i_4 - i_3) = 0 \xrightarrow{\times -1/4} 3i_1 + i_3 - 4i_4 = -2$

* KVL in Supermesh: $10 + 2i_3 - 4(i_4 - i_3) + 6(i_2 - i_1) = 0$

$\xrightarrow{\times 1/2} -3i_1 + 3i_2 + 3i_3 - 2i_4 = -5$

* Supermesh Constraint (KCL @ node a): $i_2 = i_3 + 1$

(I) $\Rightarrow \begin{cases} i_3 - 4i_4 = -14 \\ 3i_2 + 3i_3 - 2i_4 = 7 \\ i_2 - i_3 = 1 \end{cases} \xrightarrow{\text{Solve}} \begin{cases} i_2 = 3\text{ A} \\ i_3 = 2\text{ A} \\ i_4 = 4\text{ A} \end{cases} \xrightarrow{\text{Thus (I)}} \begin{cases} I_1 = i_2 - i_1 = -1\text{ A} \\ I_2 = i_1 - i_4 = 0\text{ A} \\ I_3 = i_4 - i_3 = 2\text{ A} \end{cases}$